LAB 2 — OOP Introduction

Question 1

Create a `Student` class with the following requirements:

1. Instance variables:
   - `name`
   - `id`
   - `cgpa`
2. Constructors:
   - A **default constructor**
   - A **parameterized constructor**
3. Instance method:
   - `displayInfo()` → displays student details
4. Create **at least two objects** of the `Student` class and display their information.

Sample Solution 1:

```
class Student {
    String name;
    int id;
    double cgpa;

    // Default constructor
    Student() {
        name = "Unknown";
        id = 0;
        cgpa = 0.0;
    }

    // Parameterized constructor
    Student(String name, int id, double cgpa) {
        this.name = name;
        this.id = id;
        this.cgpa = cgpa;
    }

    // Instance method
    void displayInfo() {
        System.out.println("Name: " + name);
        System.out.println("ID: " + id);
        System.out.println("CGPA: " + cgpa);
        System.out.println("--------------------");
    }
```

```
}

public class Main {
    public static void main(String[] args) {
        Student s1 = new Student();
        Student s2 = new Student("Nasif", 221001, 3.75);

        s1.displayInfo();
        s2.displayInfo();
    }
}
```

Question 2:

Create a `BankAccount` class with:

1. Instance variables:
    - `accountHolderName`
    - `balance`
2. Static variable:
    - `bankName` (same for all accounts)
3. Constructors:
    - Parameterized constructor to initialize account details
4. Instance methods:
    - `deposit(amount)`
    - `withdraw(amount)`
    - `displayBalance()`
5. Static method:
    - `showBankName()`
6. Create **two account objects** and perform deposit/withdrawal operations.

Sample Solution:

```
class BankAccount {
    String accountHolderName;
    double balance;
    static String bankName = "ABC Bank";

    // Parameterized constructor
    BankAccount(String name, double balance) {
        this.accountHolderName = name;
        this.balance = balance;
```

```
    }

    // Instance method
    void deposit(double amount) {
        balance += amount;
        System.out.println(amount + " deposited.");
    }

    void withdraw(double amount) {
        if (amount <= balance) {
            balance -= amount;
            System.out.println(amount + " withdrawn.");
        } else {
            System.out.println("Insufficient balance!");
        }
    }

    void displayBalance() {
        System.out.println("Account Holder: " + accountHolderName);
        System.out.println("Balance: " + balance);
        System.out.println("--------------------");
    }

    // Static method
    static void showBankName() {
        System.out.println("Bank Name: " + bankName);
    }
}

public class Main {
    public static void main(String[] args) {
        BankAccount.showBankName();

        BankAccount acc1 = new BankAccount("Ali", 5000);
        BankAccount acc2 = new BankAccount("Sara", 8000);

        acc1.deposit(1000);
        acc1.withdraw(2000);
        acc1.displayBalance();

        acc2.withdraw(9000);
        acc2.displayBalance();
    }
}
```

Assignment Task 1

Create an `Employee` class with:

1. Instance variables:
    - `name`
    - `salary`
2. Static variable:
    - `employeeCount` (counts total employees created)

      Constructor:
    - Increases employee count whenever a new object is created
3. Instance method:
    - `displayEmployee()`
4. Static method:
    - `displayEmployeeCount()`